Venkata Lakshmi Parimala Pasupuleti

Software Engineer · Housing & Healthcare AI · Python · AWS · Microservices · Event-Driven Systems · Agentic AI · LangGraph · Observability · CI/CD

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PROFESSIONAL SUMMARY

Software Engineer with 8 years owning housing and property management systems at scale - 7 of them inside Entrata's Affordable Housing platform, where a wrong import decision means thousands of residents receive the wrong eligibility determination. Owned the HUD income-limit pipeline end-to-end (federal rows → 180,000 domain records), event-driven RabbitMQ processing for gross-rent updates across thousands of tenant records, AWS S3 document migration with 100% PHI compliance, and 20+ full-stack production modules at 99.5% uptime with infrastructure reliability as a non-negotiable across 100K+ monthly active users. Applied the same housing compliance domain knowledge to build Leasel: an 8-agent autonomous investigation platform (LangGraph + FastAPI) that traces compliance failures across product settings, regulatory rules, historical incidents, and live system data. Built AgentProbe, an open-source AI evaluation toolkit with a CI-integrated regression gate. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

Engineering & Backend: Python, FastAPI, PHP, JavaScript, TypeScript, SQL, Bash, Docker, Microservices, Event-Driven Architecture, Distributed Systems, AWS (S3, EC2, RDS, Glue), Redis, RabbitMQ, REST APIs, Continuous Integration, CI/CD (GitHub Actions), PostgreSQL, MySQL, Oracle, Linux, Git

AI & Agents: LangGraph, LangChain, Agentic AI, Multi-Agent Orchestration, RAG Pipelines, ChromaDB, Vector Database, Agent Observability, Agent Tracing, Guardrails, HITL, Prompt Engineering, Output Validation, OCR, Document Intelligence, OpenAI API, Anthropic API, MCP, LLM Evaluation, Hallucination Detection

ML & Analytics: Machine Learning, PyTorch, scikit-learn, NLP, BERT, spaCy, PySpark, Feature Engineering, Statistical Modeling, Tableau, Power BI, Agile/Scrum

EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer Intern

- Built an agentic mission-planning system orchestrated as a LangGraph stateful graph (parse → plan → retrieve → validate → conditional routing) - delivering a risk-aware compliance strategy with autonomous routing: compliant missions to export, non-compliant ones to rejection with grounded reasons.
- Implemented a RAG layer over FAA Part 107 + operational-guidance corpus - every compliance check functions as a hypothesis test grounded in cited source regulation; engineered validation guardrails enforcing altitude limits, battery budgets, and geometric no-fly-zone checks.

George Mason University

Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate · College of Engineering and Computing

- Delivered technical workshops to 600+ students across MIS 303: Python programming (OOP, data structures, automation) and database design (ER modeling, SQL); designed Tableau and ggplot2 dashboards for faculty leadership - 40% comprehension improvement across two cohorts.
- Built multi-source prediction models (Python, R) on WHO and UK Road Safety datasets across 190+ countries - $R^2 = 0.977$ regression accuracy; applied Isolation Forest to automate data integrity checks across 3 research pipelines.

Entrata - PropTech / Affordable Housing SaaS

Sep 2018 - Aug 2024

Senior Software Engineer · Lehi, UT (Remote)

- **Owned the HUD income-limit import pipeline end-to-end for 7 years** - few thousand federal source rows expanded into ~180,000 domain records powering eligibility, rent, HAP, and tax-credit calculations for 100K+ housing units. Every release included baseline validation, reconciliation, and rollback preparation; zero compliance violations across the ownership period.
- **Built and owned the event-driven RabbitMQ processing system** for gross-rent updates across thousands of tenant records - designed retry logic and idempotency guarantees eliminating double-processing failures; owned through production incidents and annual rent-schedule changes.
- **Led the Affordable Housing module integration for the AWS S3 document migration** - inventoried all document types, validated migrated content and metadata end-to-end, and confirmed customer-facing workflows remained functional after cutover; achieved 100% PHI compliance with Libsodium-encrypted pipelines (zero security incidents).
- **Shipped 20+ full-stack production modules** (Python, PHP, PostgreSQL, JavaScript, RabbitMQ) across 120+ Agile sprints; led microservices redesign using Docker applying scalable architecture and design patterns - 50% scalability improvement, 30% deployment time reduction; eliminated 80% of manual operational workload through Python/Bash automation.
- Optimized 200+ SQL queries (30% performance gain) via execution plan analysis, indexing, and stored procedure refactoring; mentored 5+ engineers; conducted 300+ peer code reviews; built debugging observability tools reducing debugging time 40% and keeping bug count below 10 for 6+ consecutive months.

Entrata India - Pune

Jan 2017 - Sep 2018

Associate Software Engineer → Trainee

- Built property management modules (PHP, Python, PostgreSQL) for affordable housing and rental management; led AWS S3 cloud migration of client documents - resolved 90% of legacy storage constraints; income limits import covering 1.6M HUD/Tax Credit records across 5 years.

PROJECTS

Leasel: Agentic Housing Compliance Investigation Platform

Housing AI

Python · LangGraph (8-agent) · FastAPI · TF-IDF · OCR · SQLite | github.com/venkatalakshmiparimala/Lease_complaince_raq

Built to solve a real housing compliance investigation problem - when a resident appeared ineligible, the root cause (a system setting changed by another team) was invisible to anyone looking only at the compliance module. Leasel applies the same Affordable Housing domain knowledge as an 8-agent autonomous pipeline: Triage → Retrieval → Configuration → Historical Incident → Structured Data → Reasoning → Validation → Human Escalation - producing ranked, evidence-backed root-cause hypotheses with conditional HITL escalation when confidence is too low. Three surfaces sharing one evidence model: Document Intelligence (OCR-fallback ingestion, citation correctness), Support Investigation (autonomous trace), and Tenant Self-Service (policy Q&A; eligibility always routes to staff). Achieved 93.6% precision, 87.9% F1 across 1,240 residential lease PDFs; CI-integrated regression gate with LLM-as-judge scoring.

AgentProbe: Open-Source AI Evaluation Toolkit

Observability · CI/CD

Python · FastAPI · LangGraph · ChromaDB · CI/CD (GitHub Actions) | github.com/venkatalakshmiparimala/AgentProbe

Fully local, zero-cost AI agent evaluation toolkit: 15-endpoint FastAPI backend, 5-metric eval engine (correctness, groundedness, latency, token efficiency, hallucination rate), CI-integrated regression gate for automated testing, and multi-provider LLM routing - used to evaluate Leasel across 1,240+ compliance test cases.

AI on FHIR: NLP-Driven Medical Records Query System

Healthcare

Python · Flask · LLMs · FHIR · React · Docker · spaCy · BERT · Embeddings | github.com/venkatalakshmiparimala/AI_on_FHIR

Reduced manual clinical retrieval time ~60% by building a Flask + React NLP platform translating natural-language queries (EN/ES) into FHIR API calls using LLMs, BERT, and embeddings; enforced HIPAA-compliant RBAC across structured medical records.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017